Include the Following Sections

1. Project Overview

The project aims to develop an under-250 g drone using ArduPilot and MavLink, connected to cloud infrastructure for autonomous navigation and emergency response. The MVP will provide DigiCenter with a cost-efficient, modifiable, and reliable platform for testing and further developing drone-assisted rescue and medical response solutions.

2. Functional Requirements

Minimally functionable drone:

ArduPilot flight control can fly the drone in Loiter Mode, therefore all the systems and components work without failure.

Remote control has back and forth MavLink with a drone. It is also connected to the cloud, so it can overwrite control commands of remote control when it is idle.

3. Non-Functional Requirements

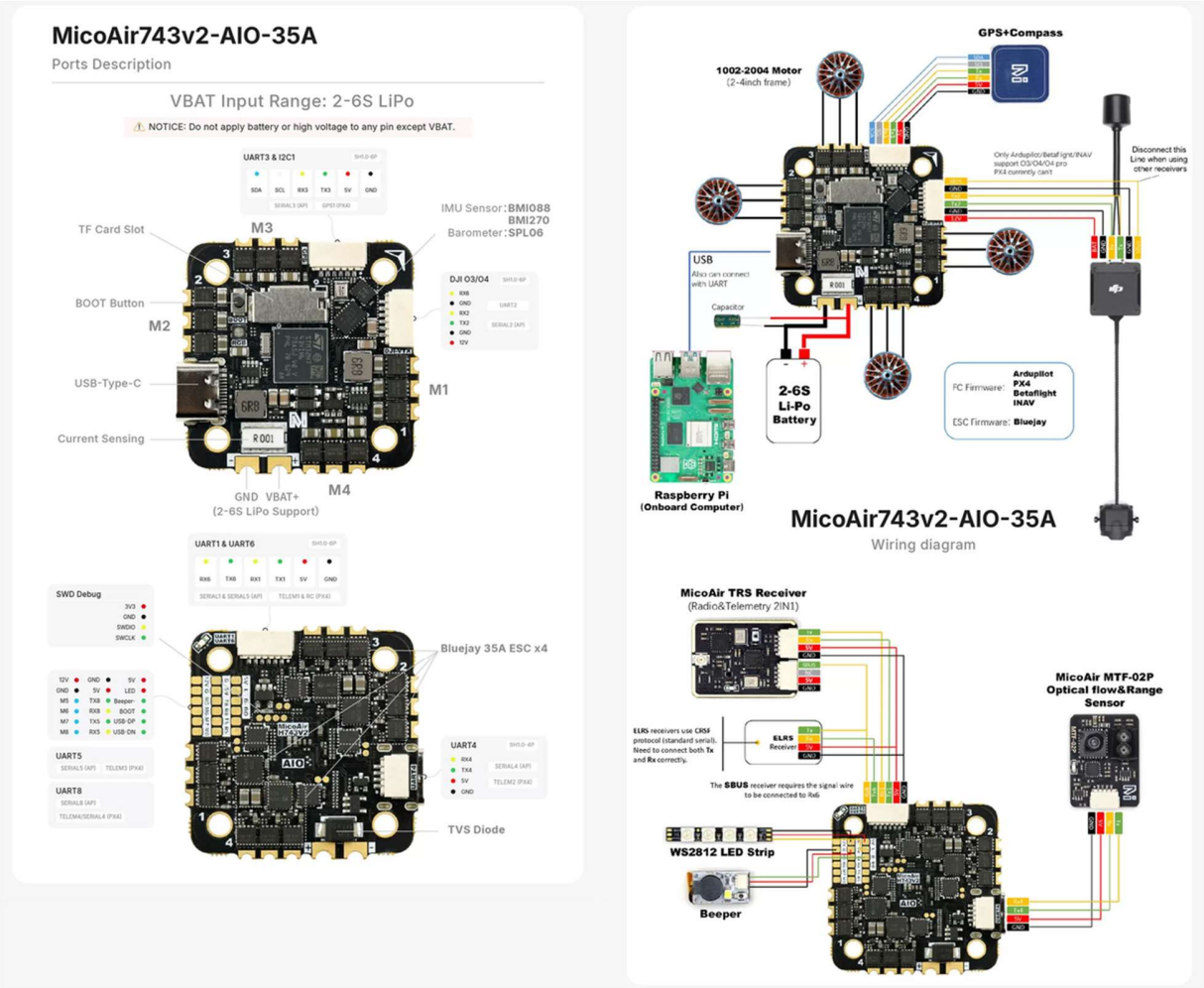
Robust design that can withstand accidental falls and hits, battery is safely wrapped and secured from punctions.

4. MVP Definition

Drone, that is under 250 g with ArduPilot and MavLink based controls and integrated communication link with cloud-based service for RC to enable autonomous navigation.

5. System Architecture

Architecture is primarily built around Flight Control which is in our case a ESC as well. Motors, battery, Video camera + dedicated RF link, ELRS radio, GPS system, Additional sensors. Remote Control.



6. Technology Selection

We selected Building drone around ArduPilot based FC, because it is open source software and hardware: there are nearly no limit of adding new features to the final product, which was a limitation for DJI drone.

7. Project Management Plan

Team roles in this project are purely formal as it is 2 people team.

8. GitHub Repository

https://github.com/Vartioee/IoT-Project-2

9. Risk Assessment

Components are not ordered on time, components are chosen incorrectly, components fail to be under 250 g.

10. Development Plan

MVP Milestone: Drone is flying with Remote Control

Testing Milestone: Cloud system is connected to drone via RC and can operate drone.

Final Integration Milestone: Drone can perform complex routing to GPS coordinates received from cloud service under.

Final Demonstration: Prove of concept ready to be scaled up.